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DETECTION AND PREDICTION OF PLUVIAL FLOOD USING MACHINE LEARNING TECHNIQUES

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ABSTRACT

The periodical occurrence of emergency situations represents an important issue for mankind. Over the years, the world at large has experienced multiple misadventures both natural and man-made. A recent report showed that flood have affected more individuals than any other category of disaster in the 21st century with the highest percentage of 43% of all disaster events in 2019 and Africa been the second vulnerable continent after Asia. Handling flood risk with the intention of safety and comfort of the citizens as well as saving their environment is one of the major responsibilities of the leadership in each country especially in flood prone areas. Machine learning predictive analytic applications can improve the risk management. So, it is highly important to devise a scientific method for flood risk reduction since it cannot be eradicated. The paper proposes a pluvial flood detection and prediction system based on machine learning techniques. The proposed model will employ a fuzzy rule-based classification to appraise the performance of the machine learning algorithm on pluvial flood conditioning variables.

Keywords: Machine Learning, Pluvial Flood, Fuzzy Rule-Based, Prediction, Detection.

1.0 INTRODUCTION

Machine learning is one of the underlying components of modern artificial intelligence and describes the ability for computers to essentially program themselves by making their own predictions and decisions based on the data they have already come across. Ordinarily it takes human intelligence in quantitative to perform the task of predicting flood but machine learning has the ability to emulate the same logical connection, such that the experts in this task are freed from this time-consuming chore and focus on other things. Here, machine learning would be helping disaster managers add true value to their services and reducing the threat to human's life and the world's economy due to the change in climate, rapid urbanization and high rate of socio-economic development that has naturally increased the risk associated with flooding.

Over the years, the world at large has experienced multiple misadventures both natural and man-made. The natural emergencies include earthquakes, tornadoes, flood, tsunamis, and landslides while the man-made include airplane crash, nuclear accidents and wars. According to International Disaster Database [4], in 2019 it was recorded that 396 natural emergency incidents occurred with 11, 755 individuals dying, above 95 million individuals influenced and approximately 130 billion dollars in economic misfortunes over the globe. Floods and storms accounted for 68% of the total number of losses, from aforesaid gravity data, it indispensable to take some stratagems to ease such kinds of harms and effects on mankind's regular and socio-economic advancement [5]. Handling flood risk with the intention of safety and comfort of the citizens as well as saving their environemnts is one of the major responsibilities of each country leadership especially in flood prone areas.

One category of natural hazards, pluvial floods with alternative terms which include "Sheet", "Street", or "overland flooding" [16] that is the place high-force precipitation surpasses the limit of urban drainage framework. This flood type is exclusive to urban areas [17]. Pluvial flood was problematized on account of land use charge, geomorphology, drainage abilities failure, pitiable urban planning and also perceived with regard to the measure of its effects which in actual fact appear to show a positive relationship with the huge amount of human inhabitants and infrastructural advancement in the cities. Literature has established that developed area seems to be the dilemma of extensive flooding effects which are equally essential to several nation's sustained economic growth [1][3][9][12]. At large, flood management has advanced from flood control tactics to flood risk administration. Governments, are therefore under burden to advance a consistent and precise maps of flood susceptible area, advance strategy for maintainable risk administration in flood with emphasis on preparation, avoidance and fortification though this is not a try to eradicate flood risk but its goal to alleviate it. Flood are foreseen to transpire more sternly in urban areas across the globe [10], findings have shown that there exists three key steps in pluvial flood risk management namely, flood planning mitigation, responsive measures and recovery [19][10] which are subject generally to two approaches non-structural and structural ways. Structural approach are measures which involves physical construction such as flood defence, catchment and coastal defences [6][7] and are based on engineered solutions, such as river dike that modifies the river flow, dams or levees [10]. The basic idea comprise of storing, averting and confinement of floods to reduce risk. On the other hands, several methods were developed by various researches in non-structural measures which include enlightening, emergency services, reporting, building codes, cautioning and forescasting, flood indemnity, assessing methods, health and social methods, land use development and public contribution [10]. In recent years, methods of mitigating and preventing flood disasters have moved from defensing approach to management approach, this is based on the comprehensive risk assessment findings and cost with benefit analysis. Some studies have indicated the need for a total set of exercises for flood risk administration through non-structural or strucural methods some time recently and after the risk despite the fact that many action have been initiated as an attempt to mitigate flood risk including machine learning [11].

In recent years, a number of approaches as reported in literatures has been in use to detect and predict flood vulnerable zones such as multi-criteria method to define the operative deployment of geospatial systems for reducing flood threat, mathematical index for large-scale analysis of flood suscepti-bility, social research information integration into flood risk analyses, remote sensing, multi-criteria decision approach in evaluating flood susceptibility, multidimensional model for urban flood risk assessment, satellite imageries in identifying areas susceptible to flooding. Recent methods include machine learning means such as artificial neural network, linear regression, support vector machine, and many of this can predict areas susceptible to flooding and map flood susceptibly areas. Most of these methods unveil certain weakness in which underscore the need for this study to employ a fuzzy rule-based classification method as dimensionality reduction algorithm for feature selection technique on various pluvial flood conditioning variables alongside machine learning algorithm. It was largely documented that, at present there is a need to improve on strategies and policies that can diminish the chance of disaster besides the method to moderate such dangers is to make certain that appropriate models remain set in motion [14].

In this paper, we proposed a model to detect and predict pluvial flood susceptibilities based on the categorization. The existing models that have used machine learning algorithms in flood prediction will be leveraged on in the design of the proposed model. The proposed model will use a fuzzy rule-based classification method as dimensionality reduction algorithm for dataset feature selection.

2.0 RELATED WORK

A real time multistep-ahead forecast model developed by [2] adopted two dynamic–Elman Neural Network, Non-linear autoregressive network and one static–back propagation neural network (three artificial neural networks) with statistical techniques (correlation analysis and Gamma test) to make water level prediction for urban pluvial flood administration. It has the ability to unravel the problematic of lasting dependencies in a time series. It effectively discovers the continual dependencies through the recursive solutions and alleviate the instability problem in the solutions but the time frame of rainfall affects the reliability and accuracy in the forecast model. In the same vein, [20] created region flood susceptibility model using fuzzy logic, weight linear combination and multi-criteria ranking methods. This was used through geographic information system to produce susceptibility model. The use of advanced optimization methods has improved the susceptibility map. Also, improved the interdependencies of flood generation variables. The assumption of the natural features being constant may affect the accuracy of the model. In addition, [15] proposed an integrated multi-parametric approach. The approach created an easy to read pluvial flood vulnerability map, using analytical hierarchy process as multi-criteria analysis instrument in creating a three-level hierarchical structure that represent pluvial flood vulnerability and geographic information system used in forming the criteria maps. The integration of AHP/GIS offers a way out to multi-parametric and multi-dimensional composite processes. It shows it’s possible practicality to broader applications where multi-parametric geospatial examination are involved. This approach was less expensive, and easy to use since the AHP assessment indices and arithmetical results can be used as a locus point but there is difficulty in standardizing the dataset for flood risk evaluation.

Semi-supervised Machine Learning Model was created by [10] by using a weakly labelled support vector machine model as an enhanced support vector machine developed for urban flood susceptibility. It can efficiently manage weakly labelled information in huge datasets and suitable for binary weakly labelled problems but the use of both uncertain labelled records and enormous unlabeled data can advance the accuracy of the model. Also, [13] designed a water level model. The use of machine learning approaches for open data using Bayesian linear model for predicting flood peak in cities. The design has the ability to handle complex task. It can be used as a short period warning system but a large error occurred when escalating the period of time of ahead of 12 hours but the model was limited by time and resources and an alteration in data set can create a giant strides in the model. In the like manner, [8] conducted a data mining model using frequency ratio and logistic ratio models as data mining techniques with geographic information system tools. This was used in generating susceptibility map to correlate amid flood information and related conditioning variables. The model was advantageous in clarifying the mechanism in the middle of flood occasions and related factors but there were difficulty in acquiring data set and the model validation was affected due to lack of data. [18] proposed a rainfall-runoff forecasting model. The model put forward an interactive and helpful structure (data driven –Random Forest and a multilayer perception Artificial Neural Networks) in refining the monitoring and administration of urban flooding using historical rainfall and runoff data. It shows promise in the absence of hydraulic model. To improve the prediction of pluvial flows, test of different data transformation technique will be needed. This justifies the reason why a dimensionality reduction feature selection approach will be applied as conditioning variable evaluator with the aim of obtaining optimum set of conditioning factors for risk assessment in flood.

3.0 PROPOSED PLUVIAL FLOOD DETECTION AND PREDICTION SYSTEM

The gaps identified in the existing models aided the guide to the improvement of a highly predictive conditioning variables subset for the proposed model. The proposed model will be aimed at evaluating the dataset (conditioning variables) using fuzzy rule based rule (fuzzy logic technique) and with five machine learning algorithms. The concept of fuzzy logic is a computerised thinking technique which can imitate compound human ideas. The strength lies in the addition of logics (Boolean) to a fuzzy set of partial truths, which outputs are continually explained within 1 and 0. It comprises three main operations. Firstly, is the fuzzification, which draws an input example to a membership importance using the membership function. This was followed by inference, in this section, the fuzzified data will be deduced and analysed considering some set of fuzzy rules as detailed in Table 1. Lastly, defuzzification was used to assign the analysed output variables with the precise decision.

The model is proposed to have four components as shown in Figure 1, namely:

(i) input stage, which is the data collection phase where the conditioning variables are supplied into the model based on the categorization
(ii) the data storage stage where the dataset are stored for processing
(iii) model components which comprises of the model building and prediction components, this contained the python code for the selected machine learning model to predict pluvial flood susceptibilities.
(iv) The output stage which displays the pluvial flood susceptibilities according to predicted multi-classification.

Table 1: Resulting fuzzy rules applied during inference

Rule No	Fuzzy Rules	
	Antecedent (IF)	**Consequence (THEN)**
R1	Rainfall is 1 and Drainage is 1	No Flood
R2	Rainfall is 1 and Drainage is 2	No Flood
R3	Rainfall is 1 and Drainage is 3	No Flood
R4	Rainfall is 1 and Drainage is 4	No Flood
R5	Rainfall is 1 and Drainage is 5	No Flood

R6	Rainfall is 2 and Drainage is 1	Low Level Flood
R7	Rainfall is 2 and Drainage is 2	Low Level Flood
R8	Rainfall is 2 and Drainage is 3	Low Level Flood
R9	Rainfall is 2 and Drainage is 4	Low Level Flood
R10	Rainfall is 2 and Drainage is 5	Low Level Flood
R11	Rainfall is 3 and Drainage is 1	Moderate Level Flood
R12	Rainfall is 3 and Drainage is 2	Moderate Level Flood
R13	Rainfall is 3 and Drainage is 3	Moderate Level Flood
R14	Rainfall is 3 and Drainage is 4	Moderate Level Flood
R15	Rainfall is 3 and Drainage is 5	Moderate Level Flood
R16	Rainfall is 4 and Drainage is 1	High Level Flood
R17	Rainfall is 4 and Drainage is 2	High Level Flood
R18	Rainfall is 4 and Drainage is 3	High Level Flood
R19	Rainfall is 4 and Drainage is 4	High Level Flood
R20	Rainfall is 4 and Drainage is 5	High Level Flood
R21	Rainfall is 5 and Drainage is 1	Very High Level Flood
R22	Rainfall is 5 and Drainage is 2	Very High Level Flood
R23	Rainfall is 5 and Drainage is 3	Very High Level Flood
R24	Rainfall is 5 and Drainage is 4	Very High Level Flood
R25	Rainfall is 5 and Drainage is 5	Very High Level Flood

4.0 FUTURE WORK

The proposed system is yet to be implemented hence future work will include full system implementation and improvement of system accuracy through the deployment of five-machine learning algorithm namely K-Nearest Neighbour, Random Forest, Naïve Bayes, Artificial Neural Network and Classification and Regression Tree. Furthermore, four measures of performance which comprised the sensitivity, specificity, accuracy percentages and precision shall be used in evaluating the predictive capability of the proposed model as well as producing a pluvial flood susceptibility map with categorization.

5.0 CONCLUSION

This paper discusses the method of detecting and predicting pluvial flood susceptibilities using machine learning techniques and as well proposes a model for predicting pluvial flood using five-machine learning algorithms. The proposed model will use a fuzzy rule-based classification technique as dimensionality reduction algorithm for feature selection and evaluate the machine learning algorithm performance on each set of conditioning variables based on the conditioning variable predictive power that is whether it is high or low.

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